

**Final answers** (record here — graded on the answer; show your work):

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_  
8. \_\_\_\_\_ 9. \_\_\_\_\_ 10. \_\_\_\_\_

## Even, Odd, or Neither

**Example (Determine even, odd, or neither):**

$f(x) = x^4 - x^2$ :  $f(-x) = x^4 - x^2 = f(x)$ , so it is **even**.

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**Directions:** Compute  $f(-x)$  and decide: even, odd, or neither. Show the substitution.

1.  $f(x) = x^2 + 1$

6.  $f(x) = x^4 - 3x^2$

2.  $f(x) = x^3 - x$

7.  $f(x) = x^5$

3.  $f(x) = x^3 + 2$

8.  $f(x) = x^2 + x$

4.  $f(x) = |x|$

9.  $f(x) = \frac{1}{x}$

5.  $f(x) = 2x$

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**Challenge** (same sheet):

10.  $f(x) = x^3 + x^2$  — test  $f(-x)$  and classify; explain why a single  $x^2$  term breaks odd symmetry.

## Answer Key — fully worked

**Procedure:** Replace every  $x$  with  $-x$  and simplify; If  $f(-x) = f(x)$  it's even (y-axis symmetry); If  $f(-x) = -f(x)$  it's odd (origin symmetry); Otherwise it's neither.

1.  $f(-x) = x^2 + 1 = f(x)$ : even
2.  $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$ : odd
3.  $f(-x) = -x^3 + 2$ : neither
4.  $f(-x) = |-x| = |x| = f(x)$ : even
5.  $f(-x) = -2x = -f(x)$ : odd
6.  $f(-x) = x^4 - 3x^2 = f(x)$ : even (all even powers)
7.  $f(-x) = -x^5 = -f(x)$ : odd
8.  $f(-x) = x^2 - x$ : neither
9.  $f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$ : odd
10.  $f(-x) = -x^3 + x^2$ : neither — the even  $x^2$  term stays + while  $x^3$  flips, so it can't equal  $\pm f(x)$